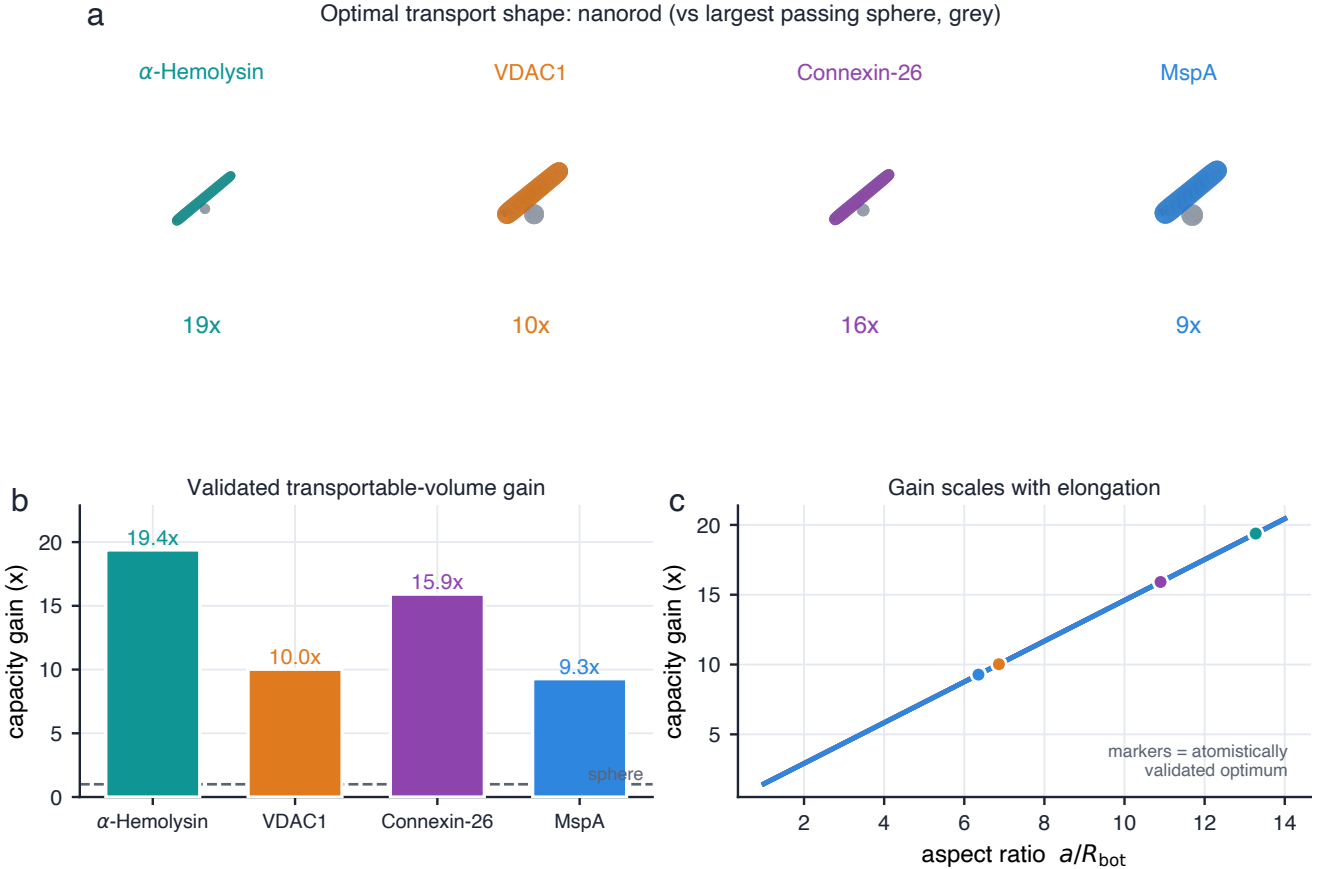

a
Optimal transport shape: nanorod (vs largest passing sphere, grey)
α-Hemolysin
VDAC1
Connexin-26
MspA
19x
10x
16x
9x
b
Validated transportable-volume gain
capacity gain (x)
19.4x
10.0x
15.9x
9.3x
sphere
α-Hemolysin
VDAC1
Connexin-26
MspA
c
Gain scales with elongation
capacity gain (x)
markers = atomistically validated optimum
aspect ratio a/R_bot